

Missing value

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Fill the missing value manually

```
#importing the dataset
library(readr)
agri_dataset <- read_csv('G:/Lesson/Done/Research/R Language/R-Language/Basic/agri_dataset.csv')
```

```
Rows: 20 Columns: 5
-- Column specification -----
Delimiter: ","
chr (1): Region
dbl (4): Crop_Yield, Fertilizer, Irrigation, Soil_pH

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#find missing value
is.na(agri_dataset$Crop_Yield)
```

```
[1] FALSE FALSE TRUE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE
[13] FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
#count the missing value
sum(is.na(agri_dataset$Crop_Yield))
```

```
[1] 3
```

```
#omit missing value -> omit all of the missing value and just give the clear dataset
na.omit(agri_dataset)
```

```
# A tibble: 13 x 5
```

	Crop_Yield	Fertilizer	Irrigation	Soil_pH	Region
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	2500	100	200	6.5	A
2	2700	110	210	6.7	B
3	3100	140	250	6.9	B
4	2800	100	230	6.6	A
5	3000	120	250	6.8	C
6	3300	130	260	7	A
7	3600	150	280	7.1	C
8	3700	100	210	6.5	A
9	3900	120	230	6.7	C
10	4000	130	240	6.8	A
11	4200	150	260	7	C
12	4300	100	270	6.9	A
13	4400	110	280	7.1	B

```
#fill the missing value with mean
mean_of_yield <- mean(agri_dataset$Crop_Yield, na.rm=TRUE)
agri_dataset[is.na(agri_dataset$Crop_Yield), "Crop_Yield"] <- mean_of_yield
print(agri_dataset)
```

```
# A tibble: 20 x 5
```

	Crop_Yield	Fertilizer	Irrigation	Soil_pH	Region
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	2500	100	200	6.5	A
2	2700	110	210	6.7	B
3	3476.	120	220	6.8	C
4	2900	130	NA	7	A
5	3100	140	250	6.9	B
6	3200	150	260	NA	C
7	2800	100	230	6.6	A

8	3476.	110	240	6.7 B
9	3000	120	250	6.8 C
10	3300	130	260	7 A
11	3400	140	NA	6.9 B
12	3600	150	280	7.1 C
13	3700	100	210	6.5 A
14	3476.	110	220	6.6 B
15	3900	120	230	6.7 C
16	4000	130	240	6.8 A
17	4100	140	250	NA B
18	4200	150	260	7 C
19	4300	100	270	6.9 A
20	4400	110	280	7.1 B

Fill the missing value with package

```
#Visualizing the missing value
library(VIM)
```

Loading required package: colorspace

Loading required package: grid

VIM is ready to use.

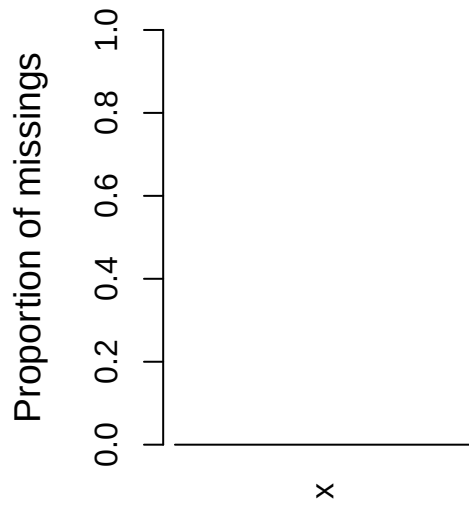
Suggestions and bug-reports can be submitted at: <https://github.com/statistikat/VIM/issues>

Attaching package: 'VIM'

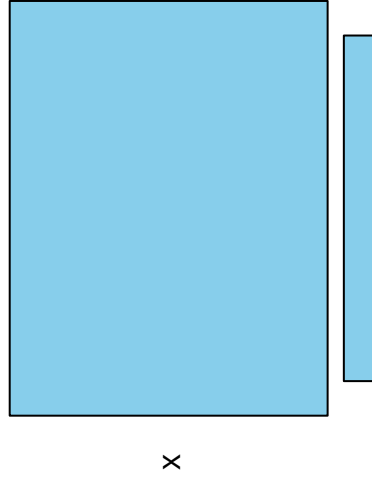
The following object is masked from 'package:datasets':

sleep

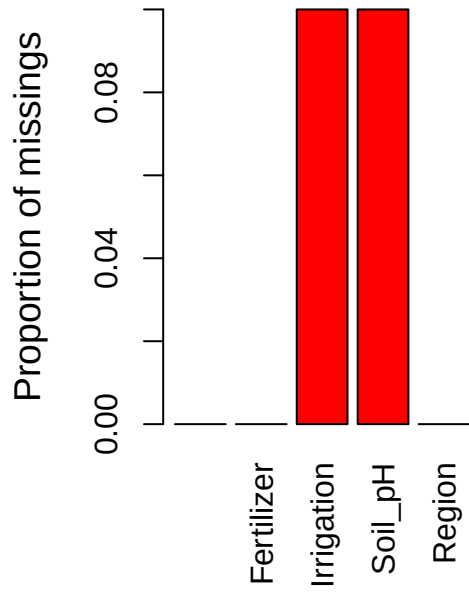
```
aggr(agri_dataset)
```



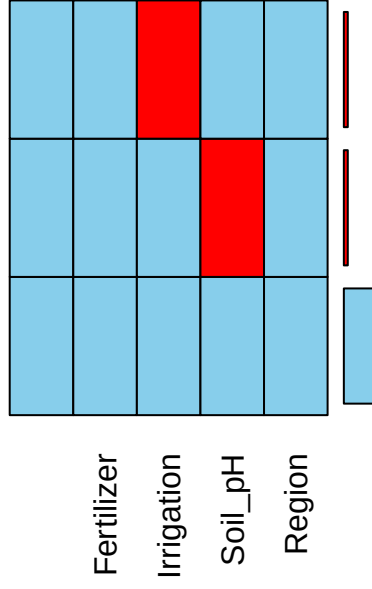
Combinations



```
aggr(agri_dataset$Crop_Yield)
```



Combinations



```
#Fill the missing value
library(mice)
```

Attaching package: 'mice'

The following object is masked from 'package:stats':

filter

The following objects are masked from 'package:base':

cbind, rbind

```
imp <- mice(agri_dataset)
```

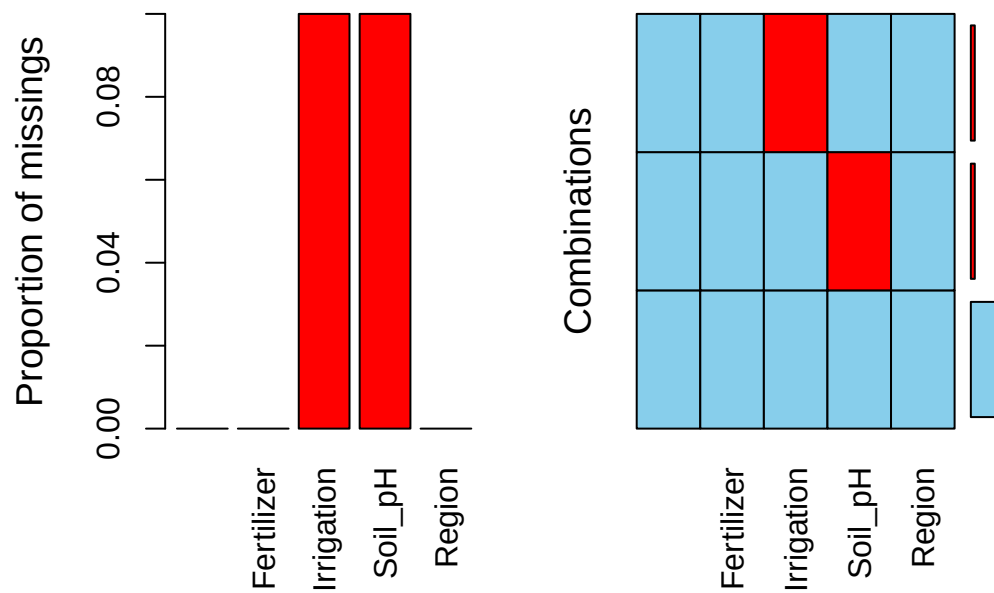
```
iter imp variable
1 1 Irrigation Soil_pH
1 2 Irrigation Soil_pH
1 3 Irrigation Soil_pH
1 4 Irrigation Soil_pH
1 5 Irrigation Soil_pH
2 1 Irrigation Soil_pH
2 2 Irrigation Soil_pH
2 3 Irrigation Soil_pH
2 4 Irrigation Soil_pH
2 5 Irrigation Soil_pH
3 1 Irrigation Soil_pH
3 2 Irrigation Soil_pH
3 3 Irrigation Soil_pH
3 4 Irrigation Soil_pH
3 5 Irrigation Soil_pH
4 1 Irrigation Soil_pH
4 2 Irrigation Soil_pH
4 3 Irrigation Soil_pH
4 4 Irrigation Soil_pH
4 5 Irrigation Soil_pH
5 1 Irrigation Soil_pH
5 2 Irrigation Soil_pH
```

```
5 3 Irrigation Soil_pH
5 4 Irrigation Soil_pH
5 5 Irrigation Soil_pH
```

Warning: Number of logged events: 1

```
clean_agri_dataset <- complete(imp)

#Visualizing the dataset after filling the missing value
aggr(agri_dataset)
```



```
aggr(clean_agri_dataset)
```

